

The induced-forest number of the Higman–Sims graph

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Abstract

The largest induced forest in the Higman–Sims graph has 26 vertices. We give an explicit induced tree attaining this value and exclude every 27-vertex induced forest, including disconnected forests. The upper bound uses the rank-22 positive semidefinite matrix $2I - A + J/5$, where A is the adjacency matrix. An eigenvector-support argument reduces a hypothetical 27-vertex forest to four small configurations, each contradicted by an exact rank computation supplied explicitly. Consequently the minimum feedback vertex set has size 74. Bibliographic priority of the exact value has not been established.

1 Statement and graph model

For a finite simple graph G , write

$$f(G) = \max\{|S| : S \subseteq V(G), G[S] \text{ is a forest}\}.$$

Forests may be disconnected and may contain isolated vertices. A feedback vertex set is a set whose deletion leaves a forest, so its minimum cardinality is $|V(G)| - f(G)$.

Theorem 1. *For the Higman–Sims graph H , one has $f(H) = 26$. Equivalently, the minimum cardinality of a feedback vertex set of H is 74.*

We use the Witt-design model of the Higman–Sims graph; the historical construction is associated with Higman and Sims [1]. Here is a precise labeling, also specifying the finite input to our checks. Over $\mathbb{F}_2$, put

$$g(x) = x^{11} + x^9 + x^7 + x^6 + x^5 + x + 1.$$

For each polynomial $m(x)$ of degree at most 11, take the 23 coefficients of $m(x)g(x)$ in degrees $0, \dots, 22$ and append their sum as coordinate 23. Coordinates are numbered from zero in this construction. Among the resulting 24-bit words, retain those of weight eight with coordinates 22 and 23 both equal to one. Their supports on coordinates $0, \dots, 21$, with each coordinate increased by one, form the 77 blocks of the $S(3, 6, 22)$ design. Order these six-element blocks lexicographically.

Vertex 1 is ∞ ; point i is vertex $i + 1$ for $1 \leq i \leq 22$; the ordered blocks are vertices $24, \dots, 100$. Join ∞ to every point, a point to every block containing it, and two blocks when they are disjoint. There are no other edges. The resulting graph has parameters $(100, 22, 0, 6)$: every vertex has 22 neighbors, adjacent vertices have no common neighbor, and distinct nonadjacent vertices have six. The supplementary program `verify.py` constructs this model, checks all 4096 words and all 1540 design triples, and checks these graph parameters and the witness below by exact integer operations.

For context, an elementary starting interval is $23 \leq f(H) \leq 33$. The lower bound comes from ∞ and the 22 points. The graph identities below give adjacency eigenvalues 22, 2, -8 ; hence for an s -vertex set with e induced edges,

$$2e \geq \frac{22s^2}{100} - 8 \left(s - \frac{s^2}{100} \right) = \frac{3s^2}{10} - 8s.$$

If the set induces a nonempty forest, $e \leq s - 1$, yielding $3s^2 - 100s + 20 \leq 0$ and therefore $s \leq 33$. Theorem 1 replaces this interval by an exact value through the rank and support argument below. This interval is a directly derived baseline, not an assertion about the best published bounds.

As adjacent work on the same graph invariant, Zeng [2] studies induced-forest bounds in even-hole-free graphs. That reference is included for subject context; no result from it is used here. The available literature assessment does not establish priority, the nearest published exact result, or whether this numerical value was previously recorded. We make no such claims, and do not classify all maximizing vertex sets.

2 An explicit 26-vertex induced tree

In the labeling above, take vertex 61, the five hubs in the left column of Table 1, and their twenty listed leaves. Each hub is adjacent to 61 and to its four leaves; there are no other edges among these 26 vertices. These assertions follow by applying the stated incidence and disjointness rules, and are checked exhaustively over the selected vertex pairs by `verify.py`.

Hub adjacent to 61	Its four leaves
5	25, 50, 62, 79
18	38, 47, 58, 100
53	6, 8, 32, 90
60	15, 21, 42, 67
97	17, 23, 24, 56

Table 1: The complete edge description of a 26-vertex induced tree.

Thus the induced graph is connected with 25 edges, and is a tree. This proves $f(H) \geq 26$.

3 A rank constraint and tree eigenspaces

Let A be the adjacency matrix of H , and let J be its all-ones matrix. The graph parameters give

$$AJ = 22J, \quad A^2 = 16I - 6A + 6J. \quad (1)$$

In particular A acts as 22 on the constants and has eigenvalues in $\{2, -8\}$ on their orthogonal complement, as used above. Define

$$N = 2I - A + \frac{1}{5}J.$$

Equation (1), together with $J^2 = 100J$, gives $N^2 = 10N$. Since N is real symmetric and $\text{tr } N = 220$, it is positive semidefinite of rank 22. For every induced subgraph F it follows that

$$M = 2I - A_F + \frac{1}{5}J_F \succeq 0, \quad \text{rank } M \leq 22. \quad (2)$$

All matrix identities here can also be verified entry by entry from the explicit graph model. The proof will exclude any forest F of order 27 satisfying (2).

Lemma 2. *If a tree has a λ -eigenvector nonzero at every vertex, then λ is simple. If the eigenvalue 2 has multiplicity m in an n -vertex forest, then $5m \leq n$.*

Proof. Let x be the nonvanishing eigenvector and y any other λ -eigenvector. On an oriented edge uv put $w_{uv} = x_u y_v - y_u x_v$. The eigenvector equations imply $\sum_{v \sim u} w_{uv} = 0$ at every vertex. A divergence-free antisymmetric flow on a tree vanishes: remove a leaf, whose incident flow must be zero, and continue. Therefore y_v/x_v is constant along each edge and hence on the tree, proving simplicity.

For the second assertion let E be the 2-eigenspace, and let U consist of the vertices at which some vector of E is nonzero. If $m = 0$ there is nothing to prove. Otherwise choose $x \in E$ nonzero on all of U ; the excluded choices lie in finitely many proper linear hyperplanes. On every component C_i of the induced forest on U , the restriction of x is a nonvanishing 2-eigenvector. By the first assertion its 2-eigenspace is one-dimensional. Restriction embeds E into the direct sum of these spaces, so $m \leq c$, where c is their number.

A tree on t vertices is bipartite, so its positive and negative eigenvalues occur in opposite pairs. Since $\text{tr}(A_{C_i}^2) = 2(t-1)$, the squares of its positive eigenvalues sum to $t-1$. The eigenvalue 2 therefore requires $t \geq 5$. Thus $n \geq |U| \geq 5c \geq 5m$. $\square$

Lemma 3. *The only trees of orders five and six with eigenvalue 2 are respectively $K_{1,4}$ and the double star with two adjacent degree-three vertices and two leaves at each. They have positive 2-eigenvectors taking value 2 at non-leaf vertices and 1 at leaves, with coordinate sums 6 and 8.*

Proof. If a_j counts the j -edge matchings of a tree on n vertices, its characteristic polynomial is

$$p(t) = \sum_j (-1)^j a_j t^{n-2j}.$$

Indeed, the determinant expansion permits only fixed points and disjoint transpositions along edges, since the graph has no cycles. At order five, $p(2) = 0$ gives $a_2 = 0$. Thus every two edges meet and the tree is a star. At order six the equation becomes $4a_2 - a_3 = 16$. A tree has at most one perfect matching, since two would have an alternating cycle. Consequently $a_3 \in \{0, 1\}$, so $a_3 = 0$ and $a_2 = 4$. As there are five edges,

$$\sum_v \binom{\deg v}{2} = \binom{5}{2} - a_2 = 6.$$

The partitions of the excess degree $\sum_v (\deg v - 1) = 4$ give the following complete list of degree sequences and corresponding sums:

sequence	(5, 1, 1, 1, 1, 1)	(4, 2, 1, 1, 1, 1)	(3, 3, 1, 1, 1, 1)	(3, 2, 2, 1, 1, 1)	(2, 2, 2, 2, 1, 1)
sum	10	7	6	5	4

Only the double-star sequence survives, and its two non-leaf vertices must be adjacent. The stated eigenvectors are checked directly at every vertex, with the stated sums. $\square$

We also use the elementary fact that a connected graph has a strictly positive eigenvector for its largest adjacency eigenvalue. For completeness, maximize the Rayleigh quotient on the unit sphere. Replacing a maximizing vector by its coordinatewise absolute value cannot decrease that quotient. The resulting nonnegative maximizing vector is an eigenvector; its eigenvector equation propagates any zero coordinate to all neighbors, and hence through the connected graph. It is therefore strictly positive.

4 Reducing an arbitrary 27-vertex forest

Suppose that F is a 27-vertex induced forest. Put $B = 2I - A_F$ and $m = \dim \ker B$. Since $M - B$ has rank one,

$$\text{rank } B \leq 23, \quad m \geq 4. \quad (3)$$

First suppose $B \succeq 0$. Each component has largest adjacency eigenvalue at most 2, and some component has eigenvalue 2 because $m > 0$. A positive eigenvector of that component, extended by zero, belongs to $\ker B$ and has nonzero coordinate sum. The kernel of a sum of positive semidefinite matrices is the intersection of their kernels. Hence

$$\ker M = \ker B \cap \mathbf{1}^\perp, \quad \text{rank } M = 28 - m.$$

Lemma 2 gives $m \leq 5$, so $\text{rank } M \geq 23$, contradicting (2).

Thus B has a negative direction. It has at most one negative eigenvalue, counted with multiplicity: a two-dimensional negative subspace would contain a nonzero vector in $\mathbf{1}^\perp$, whose M -quadratic form would also be negative. Moreover,

$$\ker B \subseteq \mathbf{1}^\perp. \quad (4)$$

Indeed, if $z \in \ker B$ had nonzero sum, a suitable multiple of z could be added to a negative vector to make its sum zero without changing its negative B -quadratic form, again contradicting $M \succeq 0$.

There is exactly one component T of F with largest adjacency eigenvalue greater than 2. Every other component has largest eigenvalue strictly less than 2: equality would supply a positive kernel vector of nonzero sum, violating (4). In particular the entire m -dimensional 2-eigenspace is supported on T . This conclusion is valid without assuming that F is connected.

Apply the support construction in Lemma 2 to this eigenspace on T . Let U be its nonzero support and $Z = V(T) \setminus U$, and let $C_1, \dots, C_c$ be the components of $T[U]$. The set Z is nonempty, since otherwise the first part of the lemma would give $m = 1$. Choose a nonvanishing 2-eigenvector u_i on each C_i . Every vector in the eigenspace is $a_i u_i$ on C_i and zero on Z . Its remaining vertex equations are

$$Ra = 0, \quad m = c - \text{rank } R. \quad (5)$$

Here R_{zi} is zero when z has no neighbor in C_i , and otherwise equals the coordinate of u_i at that neighbor. A vertex z cannot have two neighbors in C_i , since that would make a cycle. Thus every attachment gives a nonzero matrix entry. Conversely, every solution of $Ra = 0$ extends to an eigenvector, proving the dimension equality.

Connectivity gives $\text{rank } R \geq 1$, so $c \geq m + 1 \geq 5$. Each C_i has at least five vertices and $Z \neq \emptyset$. Therefore

$$5c + |Z| \leq |V(T)| \leq 27$$

forces

$$c = 5, \quad m = 4, \quad \text{rank } R = 1, \quad 1 \leq |Z| \leq 2. \quad (6)$$

Every nonzero row of R has at least two entries: use the eigenvector nonzero on all of U in its equation at that zero vertex. A single nonzero summand could not vanish. If $|Z| = 2$ and both rows were nonzero, rank one would make them proportional, with the same support of size at least two. The two zero vertices and two common support components would form a cycle after contracting those components, a contradiction. Thus in this case one zero vertex has no neighbor in U . Connectivity makes it a leaf attached to the other zero vertex, which attaches to all five components.

Lemma 3 and the vertex count now leave exactly three forms:

1. One central zero vertex joined to four copies of $K_{1,4}$ and one six-vertex double star. This uses all 27 vertices.
2. One central zero vertex joined to five copies of $K_{1,4}$, together with one isolated vertex of F .
3. One central zero vertex joined to five copies of $K_{1,4}$, with one additional leaf at that central vertex.

The central vertex attaches at exactly one vertex of each support component. At this stage all attachment positions are allowed.

5 The final four rank obstructions

Use the positive eigenvectors from Lemma 3. For component C_i , let r_i be the value at its attachment vertex and let t_i be its coordinate sum. The equation at the central zero vertex is $\sum_i r_i a_i = 0$, while (4) requires $\sum_i t_i a_i = 0$ for every solution. These two nonzero linear functionals have that containment of kernels precisely when t is a scalar multiple of r . Thus all ratios t_i/r_i must agree.

For a star the ratios are 3 at its hub and 6 at a leaf. For the double star they are 4 at a non-leaf vertex and 8 at a leaf. The mixed-size form is impossible. In each of the two five-star forms all attachments are therefore at hubs, or all are at leaves. Exactly four possibilities remain.

By symmetry and (4), the system $By = \mathbf{1}$ is solvable. Let $q = \mathbf{1}^T y$; its value does not depend on the solution, since two solutions differ by a vector in $\ker B \subseteq \mathbf{1}^\perp$. We claim that $q \neq -5$ implies $\ker M = \ker B$. Indeed, if $Mz = 0$ and $s = \mathbf{1}^T z$, then

$$Bz = -\frac{s}{5}\mathbf{1}, \quad z = -\frac{s}{5}y + k \quad (k \in \ker B), \quad \left(1 + \frac{q}{5}\right)s = 0.$$

Thus $s = 0$ and $z \in \ker B$. The reverse inclusion follows from (4). Since $m = 4$, this would force $\text{rank } M = 23$, contrary to (2).

The four exact values of q are displayed in Table 2. We derive them to make the finite computation explicit.

Attachment within each star	Additional vertex	q
Hub	Isolated	$-27/2$
Leaf	Isolated	$-27/2$
Hub	Leaf at the central vertex	-12
Leaf	Leaf at the central vertex	$-3/2$

Table 2: All remaining 27-vertex forests have $q \neq -5$ and hence rank 23.

For hub attachments, write a for the central coordinate, b_i for the i th hub coordinate, and c_i for its four equal leaf coordinates in a solution of $By = \mathbf{1}$. The leaf coordinates are equal because their equations are identical. Then

$$2c_i - b_i = 1, \quad 2b_i - 4c_i - a = 1,$$

so $a = -3$. If there is a central leaf, its coordinate is $e = (1 + a)/2 = -1$; otherwise use $e = 0$ as a bookkeeping convention. The central equation gives $\sum_i b_i = -7 - e$. Consequently the sum on the central tree, including a central leaf when present, is

$$a + \sum_{i=1}^5 (b_i + 4c_i) + e = -14 - 2e.$$

For an isolated additional vertex, its coordinate is $1/2$, giving $q = -14 + 1/2 = -27/2$. For a central leaf $e = -1$, giving $q = -12$.

For leaf attachments, write b_i for the hub, c_i for the attached leaf, and d_i for each of the other three leaves. The equations are

$$2d_i - b_i = 1, \quad 2c_i - b_i - a = 1, \quad 2b_i - c_i - 3d_i = 1.$$

They give $a = -6$. Use $e = 0$ in the isolated-vertex case, or $e = (1 + a)/2 = -5/2$ in the central-leaf case. The central equation and the attached-leaf equations imply

$$\sum_i c_i = -13 - e, \quad \sum_i b_i = -1 - 2e.$$

The coordinate sum on the central tree and any central leaf is

$$a + \sum_{i=1}^5 (b_i + c_i + 3d_i) + e = -14 - 5e.$$

This yields $q = -27/2$ after adding an isolated coordinate $1/2$, or $q = -3/2$ when $e = -5/2$. In both attachment types these systems are consistent: choose the five b_i with the specified sum and then define the remaining coordinates from their displayed equations.

None of the four sums equals -5 , so all four possibilities violate the rank bound. This excludes every induced forest of order 27. Deleting vertices from any larger induced forest leaves one of order 27. Together with Section 2, this proves Theorem 1 over all vertex subsets, including disconnected forests. The feedback vertex set assertion follows from $100 - 26 = 74$.

6 Reproducibility and assistance

The supplementary standard-library Python program `verify.py` reconstructs the graph directly from the polynomial definition. It checks the codeword weight distribution, every design triple, all graph parameters, every entry of the integer identity $(10I - 5A + J)^2 = 50(10I - 5A + J)$, and the complete induced edge set of Table 1. It also constructs the four final forests and checks their ranks and the values in Table 2 with rational Gaussian elimination. It reads no graph database and uses no floating-point arithmetic. These finite checks verify the explicit objects and calculations; the exhaustive reduction of arbitrary forests is the argument in Sections 4 and 5.

AI assistance through OpenAI Codex was used in proof development, programming, a separate mathematical review pass, and preparation of the manuscript. The tree was found during 5000 seeded greedy trials; that search was not exhaustive. Exact verification of its edge set is separate from its discovery. The mathematical review additionally reconstructed all unlabelled trees through order six and checked all 22 reduced attachment types before the coordinate-sum filter. A 300-second SAT attempt to exclude 27 vertices returned UNKNOWN; it supplies no infeasibility certificate and is not used in the proof. The separate model review is not human peer review or formal proof assistant verification.

Primary-source retrieval attempts on 8 September 2026 were unsuccessful. The historical reference below is retained from the originating problem record, and the adjacent induced-forest reference is retained from the supplied bibliography. Neither is a mathematical premise for the rank argument. Priority and a comprehensive comparison with published induced-forest or feedback-vertex-set results remain unverified.

References

- [1] D. G. Higman and C. C. Sims, *A simple group of order 44,352,000*, Mathematische Zeitschrift **105** (1968), 110–113.
- [2] Zijian Zeng, *A sharp diameter bound for induced forests in even-hole-free graphs*, Zenodo preprint, September 2026. doi:10.5281/zenodo.22294612.